

CDE Loss — SDSS

Parametric

Nonparametric

Foundation

n	Parametric												Nonparametric						Foundation				
	LinearGauss-Homo	LinearGauss-Hetero	Student-t	LogNormal-Homo	LogNormal-Hetero	Gamma-GLM	LinGauss-Homo-Ridge	LinGauss-Hetero-Ridge	Student-t-Ridge	LogNormal-Homo-Ridge	LogNormal-Hetero-Ridge	Gamma-GLM-Ridge	FlexCode-RF	BART-Homo	BART-Hetero	Flow-Spline	Quantile-Tree	CatMLP	MDN	TabPFN Native	TabPFN 2.5	RealTabPFN 2.5	TabICL Quantiles
500	-4.8 (0.1)	-2.3 (0.7)	-4.9 (0.2)	-4.4 (0.2)	-2.8 (0.2)	-4.3 (0.1)	-4.8 (0.1)	-2.3 (0.8)	-4.9 (0.2)	-4.4 (0.2)	-2.8 (0.3)	-3.1 (0.2)	-5.4 (9.3e-3)	-3.5 (3.7e-2)	-3.3 (0.2)	-5.5 (0.3)	-2.5 (0.3)	-4.5 (0.3)	-6.1 (0.5)	-9.1 (0.3)	-9.1 (0.3)	-9.0 (0.3)	-9.1 (0.4)
1K	-4.8 (0.2)	-2.8 (0.7)	-5.1 (0.4)	-4.4 (0.1)	-3.1 (0.5)	-4.3 (0.2)	-4.8 (0.2)	-3.0 (0.6)	-5.1 (0.4)	-4.4 (0.2)	-3.1 (0.5)	-3.2 (5.5e-2)	-5.8 (0.3)	-4.4 (3.6e-2)	-3.8 (0.1)	-7.3 (1.0)	-4.1 (0.3)	-5.7 (1.3)	-7.8 (0.3)	-10.0 (0.4)	-10.0 (0.4)	-10.0 (0.4)	-9.9 (0.4)
10K	-4.5 (3.2e-2)	-3.0 (0.2)	-4.8 (4.8e-2)	-3.9 (9.2e-2)	-2.2 (8.2e-2)	-3.9 (6.3e-2)	-4.5 (3.2e-2)	-3.0 (0.2)	-4.8 (4.8e-2)	-3.9 (9.1e-2)	-2.2 (9.0e-2)	-3.1 (1.2e-2)	-6.7 (2.4e-2)	-6.1 (4.0e-2)	-4.9 (9.8e-2)	-9.4 (0.6)	-6.4 (9.5e-4)	-8.3 (0.2)	-8.9 (0.6)	-10.9 (0.2)	-10.9 (0.2)	-10.9 (0.2)	-10.9 (0.1)
50K	-4.6 (2.5e-2)	-3.1 (8.1e-2)	-4.9 (5.0e-2)	-4.0 (2.9e-2)	-2.4 (0.1)	-4.0 (3.3e-2)	-4.6 (2.5e-2)	-3.1 (8.0e-2)	-4.9 (5.0e-2)	-4.0 (2.9e-2)	-2.4 (0.1)	-3.1 (2.6e-2)	-6.9 (3.3e-2)	-6.7 (2.6e-2)	-5.6 (9.4e-2)	-10.0 (0.3)	-8.0 (7.2e-2)	-9.6 (0.1)	-9.6 (0.2)	-11.0 (2.5e-2)	-11.0 (2.5e-2)	-10.9 (8.3e-3)	—
100K	-4.5 (1.6e-2)	-3.0 (7.4e-2)	-4.9 (4.0e-2)	-4.0 (1.7e-2)	-2.4 (4.1e-2)	-4.0 (1.5e-2)	-4.5 (1.6e-2)	-3.0 (7.2e-2)	-4.9 (4.0e-2)	-4.0 (1.7e-2)	-2.4 (4.0e-2)	-3.2 (9.3e-3)	-6.9 (1.5e-2)	-6.8 (2.2e-2)	-5.8 (4.3e-2)	-10.2 (0.3)	-8.4 (8.9e-2)	-10.0 (4.4e-2)	-9.2 (7.3e-2)	—	—	—	—
250K	-4.6 (3.3e-3)	-3.0 (2.6e-2)	-4.9 (9.6e-3)	-4.1 (2.5e-3)	-2.4 (2.6e-2)	-4.1 (8.6e-3)	-4.6 (3.3e-3)	-3.0 (2.6e-2)	-4.9 (9.5e-3)	-4.1 (2.4e-3)	-2.4 (2.7e-2)	-3.2 (1.3e-2)	-6.8 (1.1e-3)	-7.0 (3.8e-3)	—	-10.3 (0.1)	—	-10.4 (3.2e-2)	-9.5 (9.4e-2)	—	—	—	—
500K	-4.6 (4.9e-3)	-3.0 (1.6e-2)	-4.9 (1.1e-2)	-4.1 (2.8e-3)	-2.4 (1.2e-3)	-4.1 (8.6e-3)	-4.6 (3.3e-3)	-3.0 (2.6e-2)	-4.9 (9.5e-3)	-4.1 (2.4e-3)	-2.4 (2.7e-2)	-3.2 (1.3e-2)	-6.8 (1.1e-3)	-7.1 (6.9e-3)	—	-10.5 (9.7e-2)	—	-10.3 (3.2e-2)	-9.6 (9.4e-2)	—	—	—	—